

CyberShield AI

Helping people identify digital scams before they become victims.

TRACK: Open Innovation: Build the Next AI Startup

TEAM DETAILS

Team Name: **CipherX**

College / Institution: Lovely Professional University

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The Growing Threat of Digital Scams

Millions of users become victims of phishing, QR-code fraud, fake job offers, and malicious websites because existing cybersecurity tools are often too complex and fail to provide clear, actionable guidance.

Cyber fraud is evolving rapidly through phishing emails, fake job offers, QR-code scams, malicious websites, and social engineering attacks. Most users lack accessible AI-powered tools that can detect these threats and explain them in simple language before damage occurs.



Problem Overview

As digital services become an essential part of everyday life, cybercriminals exploit users through sophisticated phishing emails, fake job offers, malicious QR codes, fraudulent websites, and social engineering attacks.

Existing cybersecurity solutions are often too technical and provide warnings without clear explanations, leaving non-technical users vulnerable to financial fraud, identity theft, and data breaches

- **Who is affected**

Students & Job Seekers, Senior Citizens, Small Businesses, and Online Banking Users.

- **What's broken today**

Cybersecurity tools are complex for normal users

Alerts lack clear explanations

AI scams are becoming harder to detect

- **Why it matters now**

Rapid adoption of digital payments and online services.

Cyber fraud is targeting everyday users at an unprecedented scale.

AI-generated phishing and social engineering attacks are becoming increasingly difficult to detect.

- **What happens if it remains unsolved**

Financial Loss

Identity Theft

Data & Privacy Breaches

Reduced Trust in Digital Platforms

THE SOLUTION

CyberShield AI AI-Powered Digital Scam Detection & Prevention Platform

CyberShield AI is an AI-first cybersecurity assistant that detects phishing links, malicious QR codes, scam emails, SMS messages, and fraudulent websites. It explains threats in plain language, assigns a real-time risk score, and recommends the safest action before users become victims.



Proposed Solution

CyberShield AI provides a single intelligent platform where users can upload or scan suspicious digital content. The AI analyzes threats in seconds, explains the risk in simple language, and recommends safe actions such as Block, Proceed, or Report.

- **Key Features**

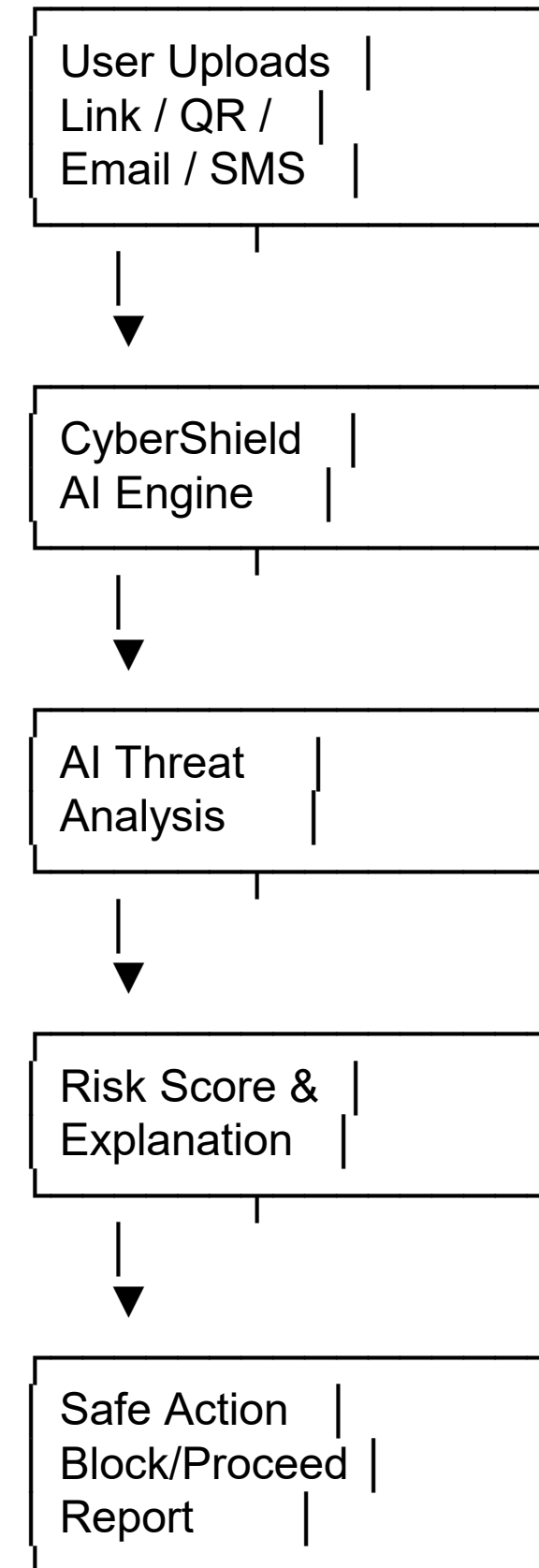
Scans links, QR codes, emails, and SMS messages to detect scams and generate an AI-powered risk assessment.

- **How It Solves the Problem**

Detects threats before users interact with malicious content and provides clear, real-time safety recommendations.

- **AI Integration**

LLMs • Machine Learning • Computer Vision • Threat Intelligence APIs



Innovation & Technology

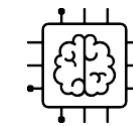


Innovation & Uniqueness

- Unlike traditional cybersecurity tools that only detect threats, CyberShield AI explains risks in simple language, analyzes multiple scam types (links, QR codes, emails, SMS, and websites) on a single platform, and provides AI-powered, real-time safety recommendations for non-technical users.
- **Unique AI Workflow**
Multi-modal analysis (Links, QR Codes, Emails & SMS)
Large Language Models (LLMs) for threat explanation
Computer Vision (OCR) for QR code analysis
Threat Intelligence APIs for URL & domain verification
AI-generated risk scoring with safe action recommendations
- Unlike Google Safe Browsing and VirusTotal, CyberShield AI not only detects suspicious content but also explains the threat in plain language, analyzes multiple scam types (links, QR codes, emails, SMS, and websites), and provides AI-powered safety recommendations tailored for non-technical users.
- **Why AI-First?**
Adapts to evolving scam techniques.
Understands context, not just keywords.
Explains threats in simple language.
Detects previously unseen phishing attempts.
Continuously improves with new data.



Technology Stack



AI / Model Layer

Google Gemini API (LLM) Phishing Detection
Model Computer Vision (OCR) AI Risk Scoring



Agents & Automation

Prompt Engineering AI Decision Engine
Explainable AI (XAI) Automated Threat Analysis



App & Backend

[React.js FastAPI (Python) Firebase
JWT Authentication]



Cloud & APIs

Google Gemini API VirusTotal API Google
Safe Browsing API Vercel & Render

From Prototype to Real-World Impact



Idea Evaluation

(Assessing Innovation, Feasibility & Real-World Impact)

Problem Relevance

Solves real-world challenges of phishing, scams, and digital fraud.

Innovation

AI-first cybersecurity assistant with real-time threat detection and simple explanations.

Feasibility

Uses existing AI models, security APIs, and scalable cloud infrastructure.

Impact

Helps users prevent fraud, improve cybersecurity awareness, and make safer digital decisions.

- **Feasibility Analysis**

Built using existing AI models, cybersecurity APIs, and cloud infrastructure

Uses proven technologies: LLMs, OCR, ML-based detection, and threat intelligence

Modular architecture enables easy development and future upgrade

- **Potential Challenges & Risks**

Evolving cyber threats may affect detection accuracy

Ensuring user data privacy and secure processing

Continuous updates required for new scam patterns

- **Strategies to Overcome Challenges**

Regular model improvement with updated threat datasets

Secure data handling and privacy-focused design

Integration with trusted security APIs for real-time intelligence